

Distortion Class Cheat Sheet

m Distortion(name, shape=None, *, label=None, **natural)

Distortion is a factory base class: the call `Distortion('ph', 0.9)` dispatches on name to a registered subclass (PHDistortion, TVaRDistortion, ...) and returns an instance. A distortion is an increasing, concave $g : [0, 1] \rightarrow [0, 1]$ with $g(0) = 0, g(1) = 1$, used to define a coherent risk measure / pricing functional. Create one by family with a named constructor (**s** `Distortion.ph(0.9)`), or via DecL `distortion <NAME> <KIND> <PARAMS>`. The following tables show all **m** methods, **s** static / class methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

name *the identity handle: the kind, or a given name*

kind *registry key for the subclass*

shape *the (single) shape parameter*

label *resolved display label*

*Named constructors, by family (each **s** static):* **s** ph

(proportional hazard), **s** wang (Wang / normal shift),

s dual (dual moment), **s** power (α on $[x_0, x_1]$),

s beta, **s** tvar (TVaR at level p), **s** bitvar

($p_0 < p_1$, weight w on p_1), **s** ccoc (constant cost of

capital from discount d), **s** mean (= TVaR at $p=0$),

s max (= TVaR at $p=1$). *Combine / generate:*

s mixture (weighted mixture), **s** minimum (pointwise

min), **s** random_distortion,

s random_distortion_ex (random wtdTVaR),

s distortions_from_params (dict from a calibration

table), **s** available_distortions (list registered

kinds), **s** decl_spec (spec dict from a DecL number

list), **s** from_spec (construct from that spec)

2. Update

None — a Distortion is a closed-form function, not a discretized object. Everything is computed on demand.

3. Moments

None — the moments of g itself (D_g statistics, and the closed forms where they exist) are tabulated in `stats_df` / `summary_df`; see Output.

4. Statistical functions

m g (the distortion $g(x)$), **m** g_inv (inverse g^{-1}),

m g_dual (dual $1 - g(1 - x)$), **m** g_prime (derivative

g' , central-difference default), **m** effective_g (resolve

the ask / bid view and the loss / payoff role to the effective

g), **m** make_q (vector of risk-adjusted probabilities q for

pricing), **s** tvar_terms (TVaR knot function

$\min(s/(1 - p), 1)$)

5. Validation

pricing_ok (the distortion yields well-defined prices),

strict_pricing (enforce strict concavity / validity),

error (numerical g vs. closed-form residual; see

`summary_df`)

6. Output dataframes

info (text summary)^[1], `summary_df`^[1] (paired $D_g, D_{g^{-1}}$

columns plus checks), `stats_df`^[1] (D_g statistics with

closed form and error where analytics exist),

`density_df`^[1] ($x, g, g^{-1}, g^*, g', \text{kusuoka}$),

`tvar_info_df` (affine pieces of a weighted-TVaR: knots,

slopes, intercepts), `decl_params`, `param_name`

7. Reinsurance

None

8. Visualization

m plot (g , dual, and diagonal)^[1], **m** plot_affine
(upper affine envelope of a wtdTVaR)

9. Risk and pricing

m price (distorted ask / bid prices of a distribution),

m calibrate (solve shape so the distorted price hits a

premium_target), **s** calibrate_set (calibrate a

whole family to one target), premium_target, assets,

a (capital level), mass, has_mass, has_mass_default

(point mass $g(0^+) > 0$), gini_p (Gini coefficient)

10. Approximations

None

11. Meta

m id (short structural ID string), med_name, long_name

(display labels), documented (listed by

`available_distortions`), **m** help,

m format_program, program, pprogram,

pprogram_html, note, tags, doc, hints (the DecL

trailer)^[2], label_map, labels, renamer, use_labels,

m min_index, **m** quick_gS, **m** quick_ra

(numba-backed numerical helpers)

Notes:

[1]: `info`, `summary_df`, `stats_df`, `density_df`, and `plot` mirror the common reporting interface shared with Aggregate and Portfolio: the same names do the analogous job on each class.

[2]: a `dist` line carries the same trailer as every other DecL statement, so a distortion can be described, tagged, and documented like an aggregate.

$g(0) = 0, g(1) = 1$, increasing and concave; dual

$g^*(x) = 1 - g(1 - x)$; price = $\int_0^\infty g(S(x)) dx$.

aggregate v.1.0.0a171

ease | confidence | precision

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